Jason Ratta

Year [redacted] Mathematics Project

DESIGNING AND CONSTRUCTING A CHOCOLATE BOX

Background

A confectionary company, NESCAD, has advertised for tenders to design and make a prototype chocolate box, which will be suitable for packaging and selling their new "Melt in Your Mouth" chocolates. The only information you have is that the chocolates must pack snugly into the box without any other packaging or inserts. It is important that the cost of the box is kept to a minimum so that means a minimum of cardboard will have to be used in the design. You have been asked to contact the company for any further information that you require. When you apply for the job, the company wants you to inform them about how much material it is going to take to make each box and the cost of the material per box.

Part One - CLARIFY THE PROBLEM

Make a list of the information that you will need to obtain from the company so that you can design the box and work out the cost of the materials to make the box. This list must be handed to the teacher by the end of the lesson and the information you have requested will be provided to you. If you find later on that you need more information, you must write a request to the teacher to get the information.

Part Two- CHOOSE A METHOD

Explain in steps, **how** you are going to work out the dimensions of your box.
Explain **how** you will come up with the cheapest design.
Explain **how** you are going to design and draw the net of the box.
Explain **how** you are going to work out how much the box costs to produce.

Part Three- USE THE METHODS YOU HAVE DESCRIBED IN PART TWO

Show all of the calculations and drawings you needed to make to design the confectionary box.
Show surface area and volume calculations and explain why they are important.
Calculate how much it will cost for the materials for each box.
Show a **scale drawing** of the **net** of your box and a **scale drawing** of how the box will look in 3 D.
Make a **real size** cardboard model of the box you have designed.

Part Four- CHECK AND VERIFY

Show evidence of checking that your finished model actually does the job it has been designed for -

- do the chocolates fit snugly into the box so that they will not move in transport? Show this in a diagram or make some model chocolates and show how they fit into the box.
- Is your design the least expensive design for the task? Show how you made sure it was.

Part Five- COMMUNICATE YOUR RESULT.

Write a letter to the company telling them who you are, why you are writing to them, about the design of your box, how much material it is going to use and how much it is going to cost.

DUE DATE: ______________________________

Costs [illegible] to make
makes 8 box. from [illegible] m^2